**IEOR 4404 Simulation** **Fall 2025**

# Practice Final Solution

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**Problem 1** To estimate $\theta$, we generate 30 independent and identically distributed values each having mean $\theta$. The sample mean is 122.85 and the sample standard deviation is 16.8.

(a) Construct a 95% confidence interval for $\theta$.

(b) Do you think your answer in part (a) is justified? Explain briefly how you justify that this is correct, or alternately what issues it has.

(c) How many additional random values do you need to generate if we want to be 95% confident that our final estimate of $\theta$ is correct to within $\pm 0.5$?

**Solution**

(a) Applying the formula from the lecture slides $[122.85-1.96\times\frac{16.8}{\sqrt{30}}, 122.85+1.96\times\frac{16.8}{\sqrt{30}}] \approx [116.83, 128.86]$

(b) The sample size is 30 which is the rule of thumb minimum to justify the use of the central limit theorem. On the other hand, the central limit theorem result is asymptotic in the sense that we want $n$ to be very large, not just larger than 30.

(c) Matching the half-width of the confidence interval gives $1.96 \times \frac{16.8}{\sqrt{n}} = 0.5 \Rightarrow n = 4337.012736$, so we need $n = 4338$ to achieve 0.5 half-width with 95% confidence. Thus we need 4308 additional samples.

**Problem 2** Messages arrive at a communication facility according to a time inhomogeneous Poisson process with intensity function $\lambda(t) = 5 + 3t$ for $0 \le t \le 5$, and $20 - 2(t-5)$ for $5 \le t \le 10$. The facility consists of one channel. An arriving message will either go to this channel if it is free, in which case the channel becomes busy, or else will be lost if the channel is busy. The amount of time $X$ that a message occupies a channel satisfies

$$P(X > s) = e^{-\sqrt{s}}, s \ge 0,$$

independent of everything else. Suppose that the system is empty at time 0. In this problem, assume your computer only has the capability to generate $Unif(0,1)$ variables.

(a) Write pseudo-codes to generate the arrivals of messages from time 0 to 10, using both the interarrival time generation approach and the conditional representation approach.

(b) We are interested in estimating $E[L]$, where $L$ is the total number of messages that are lost from time 0 to 10. Explain how you would adapt the pseudo-codes in part (a) (for both approaches therein) to simulate one replication for estimating $E[L]$. In explaining your adaptation, you should also describe how you simulate the random variable $X$.

(c) Let $T_B$ be the total amount of time in the interval $[0, 10]$ that the channel is busy. Suppose, only for this part, that instead of a time inhomogeneous Poisson process we have a time homogeneous Poisson process with rate $\lambda$ to generate the message arrivals. What is the conditional expectation $E[L \mid T_B]$?

(d) Back to using the time inhomogeneous Poisson process, let $N_B$ be the number of time intervals during $[0, 10]$ in which the channel is busy, and let those intervals be denoted by $I_1 = [a_1, b_1), I_2 = [a_2, b_2), \ldots, I_{N_B} = [a_{N_B}, b_{N_B})$. Derive an expression for the conditional expectation $E[L \mid N_B, I_1, I_2, \ldots, I_{N_B}]$ in terms of $N_B, a_1, \ldots, a_{N_B}, b_1, \ldots, b_{N_B}$.

(e) By using your answer in part (d), explain how you would modify your answer in part (b) to a conditional Monte Carlo method to estimate $E[L]$.

**Solution:**

a) We have:

$$\lambda(t) = \begin{cases} 5 + 3t, & \text{for } 0 \le t \le 5 \\ 30 - 2t, & \text{for } 5 < t \le 10 \end{cases}$$

Observe that $\lambda(t) \le 20$.

**The Interarrival Time Generation Approach:**

We start by generating a standard Poisson process with rate $\lambda_{max} = 20 \ge \lambda(t)\ \forall t$. For each incurred event at time $t$ up until time 10, we accept it with probability $\frac{\lambda(t)}{\lambda_{max}}$.

**Algorithm 1:** The Interarrival Generating Approach

```
Initialize t ← 0, A ← [];
Set λ_max ← 20;
while t < 10 do
    Generate U ~ Unif(0,1);
    Set Δt ← −log(U)/λ_max;
    t ← t + Δt;
    if t ≥ 10 then
        break;
    Compute λ(t) = { 5 + 3t     0 ≤ t ≤ 5
                   { 30 − 2t    5 < t ≤ 10 ;
    Generate V ~ Unif(0,1);
    if V ≤ λ(t)/λ_max then
        Append t to A;
return A
```

**The Conditional Representation Approach:**

We first generate $N(10) \sim \text{Pois}(20 \cdot 10)$. Suppose $N(10) = n$, we generate $U_1, \ldots, U_n \sim \text{Unif}(0,1)$ and sort them as $U_{(1)} \le \ldots \le U_{(n)}$. Then, we set $A'_i = 10 \cdot U_{(i)}$. For each $i$, we keep $A'_i$ with probability $\frac{\lambda(A'_i)}{20}$. The pseudo-code is as follows:

**Algorithm 2:** The Conditional Representation Approach

Initialize an empty set $\mathcal{A}$;
Set $p = e^{-200}$ ; `// Generate Pois`$(200)$
Initialize $F = p$, $k = 0$ ;
Simulate $U \sim \text{Unif}(0, 1)$;
**while** $U > F$ **do**
  Update $p = \frac{200 \cdot p}{k+1}$;
  Set $F = F + p$;
  Set $k = k + 1$;
Set $n = k$ ; `// This follows Pois`$(200)$
Simulate $U_1, \ldots, U_n \sim$ i.i.d. $\text{Unif}(0, 1)$;
Sort $U_{(1)}, \ldots, U_{(n)}$ in ascending order;
Set $A'_i = 10 \cdot U_{(i)}$ for $i = 1, \ldots, n$;
Generate $n$ i.i.d. $V_i \sim \text{Unif}(0, 1)$;
**for** $i = 1, \ldots, n$ **do**
  **if** $V_i \leq \frac{\lambda(A'_i)}{20}$ **then**
    Add $A'_i$ to $\mathcal{A}$;
**return** $\mathcal{A}$;

b) First, to simulate $X$, we use the inverse transform method:

$$\begin{aligned} U = F(X) &\Rightarrow U = 1 - e^{-\sqrt{X}} \\ &\Rightarrow -\log(1 - U) = \sqrt{X} \\ &\Rightarrow X = (\log(1 - U))^2 \end{aligned}$$

where $U \sim unif(0, 1)$. Again, since $U$ and $1 - U$ have the same distribution, we will use:

$$X = (\log U)^2$$

We use the following algorithm to simulate one replication of $L$:

**Algorithm 3:** Estimating $\mathbb{E}[L]$ — Number of Lost Messages

Simulate arrival times $\{t_1, t_2, \ldots, t_n\}$ using either the approaches in (a);
Initialize $L \leftarrow 0$;
Initialize $b = 0$ for busy time;
**for** $i = 1$ **to** $n$ **do**
  **if** $t_i < b$ **then**
    $L \leftarrow L + 1$ ; `// message is lost`
  **else**
    Generate $U \sim \text{Unif}(0, 1)$;
    Compute service time: $X \leftarrow (\log(1/U))^2$;
    Update busy time $b = t_i + X$;
**return** $L$ *as the total number of lost messages*

c) Assuming a homogeneous Poisson process with rate $\lambda$ and conditioning on the total time the channel is busy, denoted by $T_B$, the arrivals during this time form a homogeneous Poisson process with rate $\lambda$. Therefore, the number of lost messages is distributed as a Poisson random variable with mean $\lambda T_B$. Hence,

$$\mathbb{E}[L \mid T_B] = \lambda T_B.$$

d) Similar to the previous part, messages lost are the messages that arrived during those busy time intervals $I_i$, which follows inhomogeneous Poisson process. Thus, for any interval, expected number of arrivals is $\int_{I_i} \lambda(t)dt$. Thus,

$$\mathbb{E}[L|N_B, a_1, ..., b_{N_B}] = \sum_{i=1}^{N_B} \int_{a_i}^{b_i} \lambda(t)dt$$

We next evaluate the integral:

$$\Lambda(t) := \int_0^t \lambda(t)dt = \begin{cases} \frac{3}{2}t^2 + 5t & 0 \le t \le 5 \\ -t^2 + 30t - 87.5 & 5 < t \le 10 \end{cases}$$

Thus,

$$\mathbb{E}[L|N_B, a_1, ..., b_{N_B}] = \sum_{i=1}^{N_B} \big(\Lambda(b_i) - \Lambda(a_i)\big)$$

e) We next modify the algorithm in (b) to also keep track of the busy intervals, and evaluate the integrals for each of these intervals:

**Algorithm 4:** Estimating $\mathbb{E}[L]$ — Busy Intervals

Simulate arrival times $\{t_1, t_2, \ldots, t_n\}$ using either the approaches in (a);
Initialize $\hat{L} = 0$;
Initialize $b = 0$ for busy time;
**for** $i = 1$ **to** $n$ **do**
  **if** $t_i < b$ **then**
    ;  `// message is lost.  Nothing to be done here.`
  **else**
    Generate $U \sim \text{Unif}(0, 1)$;
    Compute service time: $X \leftarrow (\log(1/U))^2$;
    Get busy time: $a_i = t_i$ and $b_i = \min(10, t_i + X)$;
    Evaluate $\hat{L}_i = \int_{a_i}^{b_i} \lambda(t)dt = \Lambda(b_i) - \Lambda(a_i)$;
    Update $\hat{L} = \hat{L} + \hat{L}_i$
**return** $\hat{L}$ *as the estimate for expected number of messages lost*

**Problem 3** We run simulation on $Y$ to estimate our target quantity $E[Y]$.

(a) Suppose we use $Z_1$ as a control variate, and $Y$ and $Z_1$ are positively correlated. For concreteness, one replication of the control variate estimator is then $W = Y - \beta_1(Z_1 - \mu_1)$, where $\mu_1 = E[Z_1]$ is known. If we want to reduce the variance of naive Monte Carlo, should we choose $\beta_1$ to be positive, negative, or dependent on the problem? Explain your answer.

(b) Suppose now we use two control variates $Z_1$ and $Z_2$, so that both the pair $Y$ and $Z_1$, and the pair $Y$ and $Z_2$, are positively correlated. Concretely, one replication of the control variate estimator is now $W = Y - \beta_1(Z_1 - \mu_1) - \beta_2(Z_2 - \mu_2)$, where $\mu_1 = E[Z_1]$ and $\mu_2 = E[Z_2]$ are known. Provide an example where $\beta_1 > 0$ and $\beta_2 < 0$, but $Var(W) < Var(Y)$.

(c) Now suppose that $Z_1$ and $Z_2$ are independent. Can you still find an example as in part (b)? Explain briefly.

**Solution:**

a) Observe that:
$$Var(W) = \sigma_y^2 + \beta_1^2\sigma_z^2 - 2\beta_1 \text{cov}(Y, Z_1)$$
Since $\text{cov}(Y, Z_1) > 0$, we have:
$$\begin{aligned}\beta < 0 &\Rightarrow \beta_1^2\sigma_z^2 - 2\beta_1\text{cov}(Y, Z_1) > 0\\ &\Rightarrow Var(W) > \sigma_y^2\end{aligned}$$
Thus, to obtain a reduction in variance, we should always choose a positive $\beta$.

**Note:** Not every positive $\beta$ would be able to reduce variance. We will need $\beta_1^2\sigma_z^2 - 2\beta_1\text{cov}(Y, Z_1) < 0 \Rightarrow \beta < 2\text{cov}(Y, Z_1)/\sigma_z^2$ to obtain a reduction in variance.

b) We consider the case when $Z_1 = Z_2 = Y$. Therefore, both the pair $(Y, Z_1)$ and the pair $(Y, Z_2)$ are positively correlated. We can select $\beta_1 = 1$ and $\beta_2 = -0.5$. We then have
$$\begin{aligned}Var(W) &= Var(Y) + Var\left(Y - \mathbb{E}\left[Y\right]\right) + \frac{1}{4}Var\left(Y - \mathbb{E}\left[Y\right]\right)\\ &\quad + 2\left(-Cov\left(Y, Y - \mathbb{E}\left[Y\right]\right) + Cov\left(Y, 0.5\left(Y - \mathbb{E}\left[Y\right]\right)\right) - Cov\left(Y - \mathbb{E}\left[Y\right], 0.5\left(Y - \mathbb{E}\left[Y\right]\right)\right)\right)\\ &= 2\frac{1}{4}Var(Y) + 2\left(-Var(Y) + 0.5Var(Y) - 0.5Var(Y)\right)\\ &= \frac{1}{4}Var(Y) < Var(Y),\end{aligned}$$
where $Y$ is not constant.

While $\beta_1 > 0$ and $\beta_2 < 0$, we can still achieve some variance reduction from control variables that are both positively correlated with $Y$.

c) Yes, we can still find an example.

**Example 1:** Suppose $Z_1, Z_2 \overset{\text{iid}}{\sim} \mathcal{N}(0, 1)$ and define $Y = Z_1 + Z_2$. Then,
$$\text{Cov}(Y, Z_1) = \text{Cov}(Z_1 + Z_2, Z_1) = \text{Var}(Z_1) + \text{Cov}(Z_2, Z_1) = 1 + 0 = 1 > 0,$$
$$\text{Cov}(Y, Z_2) = 1 > 0 \quad \text{(similar)}.$$
Now define the control variate estimator:
$$W = Y - \frac{1}{2}Z_1 + \frac{1}{4}Z_2,$$
so that $\beta_1 = \frac{1}{2} > 0$ and $\beta_2 = -\frac{1}{4} < 0$.
We simplify:
$$W = Z_1 + Z_2 - \frac{1}{2}Z_1 + \frac{1}{4}Z_2 = \frac{1}{2}Z_1 + \frac{5}{4}Z_2.$$
Since $Z_1$ and $Z_2$ are independent and both have variance 1:
$$\text{Var}(W) = \left(\frac{1}{2}\right)^2 \text{Var}(Z_1) + \left(\frac{5}{4}\right)^2 \text{Var}(Z_2) = \frac{1}{4} + \frac{25}{16} = \frac{29}{16} \approx 1.81 < \text{Var}(Y) = 2.$$

**Example 2:** Consider the example where $Y = Z_1Z_2$, where $Z_1$ and $Z_2$ are independent Bern(0.5) random variables, and we take $Z_1$ and $Z_2$ as our two control random variables. We observe that $Y$ and $Z_1$ are positively correlated, $Y$ and $Z_2$ are positively correlated, and $Z_1$ and $Z_2$ are independent.

Since $Z_1$ and $Z_2$ follow Bern(0.5), we have $\mathrm{Var}(Z_1) = \mathrm{Var}(Z_2) = 0.25$ and $\mathrm{Cov}(Z_1, Z_2) = 0$ due to independence. Next, we compute:

$$\begin{aligned}
\mathrm{Cov}(Y, Z_1) &= \mathbb{E}[YZ_1] - \mathbb{E}[Y]\mathbb{E}[Z_1] \\
&= \mathbb{E}[Z_1^2 Z_2] - \mathbb{E}[Z_1 Z_2]\mathbb{E}[Z_1] \\
&= \mathbb{E}[Z_1^2]\mathbb{E}[Z_2] - \mathbb{E}[Z_1]\mathbb{E}[Z_2]\mathbb{E}[Z_1] \\
&= \left(\mathrm{Var}(Z_1) + \mathbb{E}[Z_1]^2\right)\mathbb{E}[Z_2] - \mathbb{E}[Z_1]^3 \\
&= \left(0.25 + 0.5^2\right) \cdot 0.5 - 0.5^3 \\
&= 0.125.
\end{aligned}$$

Similarly, we have $\mathrm{Cov}(Y, Z_2) = 0.125$. Here, we choose $\beta_1 = 0.5$ and $\beta_2 = -0.1$.

Therefore,

$$\begin{aligned}
\mathrm{Var}\left(W(\beta_1, \beta_2)\right) &= \mathrm{Var}(Y) + \beta_1^2 \mathrm{Var}(Z_1) + \beta_2^2 \mathrm{Var}(Z_2) \\
&\quad + 2\left(\beta_1\beta_2 \mathrm{Cov}(Z_1, Z_2) - \beta_1 \mathrm{Cov}(Y, Z_1) - \beta_2 \mathrm{Cov}(Y, Z_2)\right) \\
&= \mathrm{Var}(Y) + 0.25 \cdot 0.25 + 0.01 \cdot 0.25 + 2\left(-0.5 \cdot 0.125 + 0.1 \cdot 0.125\right) \\
&= \mathrm{Var}(Y) - 0.035 < \mathrm{Var}(Y).
\end{aligned}$$

**Problem 4** Suppose we have the capability to generate $Exp(1)$ variable (e.g., by the inverse transform method).

(a) We are interested in estimating $E[(X+1)^2]$, where $X$ follows a gamma distribution with parameters $\alpha = \frac{3}{2}$ and $\beta = 1$, i.e., $Gamma(\alpha, \beta)$ which has a density

$$\frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, \quad x > 0$$

Write a pseudo-code that only draws $Exp(1)$ to generate one output replication to estimate $E[(X+1)^2]$.

(b) Explain why the estimator you propose in part (a) is valid, including checking any needed conditions.

(c) Based on the estimation variances, do you think the estimator in part (a) is better than the one that directly generates $Gamma(\alpha, \beta)$? Explain with calculations.

**Solution**

a) We can use the IS method to estimate $\mathbb{E}[(X+1)^2]$ for $X \sim \Gamma(1.5, 0.5)$. We first write down $g(x)$ and $L(x) = \frac{f(x)}{\tilde{f}(x)}$:

$$g(x) = (x+1)^2, \quad L(x) = \frac{\frac{x^{0.5}e^{-x}}{\Gamma(1.5)}}{e^{-x}} = \frac{\sqrt{x}}{\Gamma(1.5)}$$

Then, the IS estimator becomes:

$$g(X)L(X) = \frac{\sqrt{X}(X+1)^2}{\Gamma(1.5)}$$

where $X \sim \exp(1)$.

b) The technical conditions are satisfied since: $\tilde{f}(x) \neq 0$ whenever $g(x)f(x) \neq 0$ (which corresponds to $x \geq 0$), or equivalently $g(x)f(x) = 0$ whenever $\tilde{f}(x) = 0$ (which corresponds to $x < 0$)

c) $Var((X+1)^2) = \mathbb{E}[(X+1)^4] - [\mathbb{E}(X+1)^2]^2$. Since both of the approaches are unbiased, $\mathbb{E}[(X+1)^2]$ is same in either cases. Thus we evaluate $\mathbb{E}[(X+1)^4]$ in both the cases. We first evaluate for the estimator directly using the Gamma variable:

$$\begin{aligned}\mathbb{E}[((X+1)^2)^2] &= \int_0^\infty (x+1)^4 \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x} dx \\ &= \frac{1}{\Gamma(1.5)} \int_0^\infty (x+1)^4 x^{0.5} e^{-x} dx =: V_\Gamma\end{aligned}$$

Next, we consider $\tilde{\mathbb{E}}[(X+1)^4]$ for the IS estimator:

$$\begin{aligned}\tilde{\mathbb{E}}\left[\left(\frac{\sqrt{X}(X+1)^2}{\Gamma(1.5)}\right)^2\right] &= \frac{1}{\Gamma(1.5)^2} \int_0^\infty x(x+1)^4 e^{-x} dx \\ &= \frac{1}{\Gamma(1.5)} \int_0^\infty \frac{x^{0.5}}{\Gamma(1.5)} (x+1)^4 x^{0.5} e^{-x} dx =: V_{IS}\end{aligned}$$

The IS estimator is better if $V_{IS} < V_\Gamma$. Otherwise, directly using gamma is better off.

Numerical evaluation reveals that the variance of the IS estimator is higher, thus directly generating the gamma variable is better here (but this conclusion is not required in the exam).